COMPSCI 682 Neural Networks: A Modern Introduction, Fall 2026

Assignment 3

This assignment is due on Wednesday, November 25 at 11:55pm ET.

You can find the starter code containing Colab notebooks in `assignment3.zip`. Please note that you can finish the homework either using Google Colab or on your local machine.

- Setup
- Goals
- Q1: Image Captioning with Transformers (25 points)
- Q2: Self-Supervised Learning for Image Classification (15 points)
- Q3: Style Transfer (10 points)
- Submitting your work

Setup

Please familiarize yourself with the recommended workflow in the "Working remotely on Google Colaboratory" section of `setup_instructions.pdf`, included in `assignment3.zip`, before starting the assignment.

In order to do the assignment, you will need to install the latest PyTorch (instructions here).

If you face difficulty re-using the Python environment used in ealier assignments, we suggest that you create a fresh one. This involves only a few simple steps (using conda and pytorch cpu for the example):

```
conda remove --name cs682 --all # delete old environment if there is one
conda create --name cs682 python=3.12
conda activate cs682
pip3 install -r requirements.txt
pip3 install torch torchvision --index-url https://download.pytorch.org/whl/cpu
# cpu-only pytorch for a linux machine
```

Note. Ensure you are periodically saving your notebook (`File -> Save`) so that you don't lose your progress if you step away from the assignment and the Colab VM disconnects.

Once you have completed all Colab notebooks, proceed to the submission instructions.

Goals

In this assignment, you will implement a transformer model and apply them to image captioning on the COCO dataset. Then you will be introduced to self-supervised learning to automatically learn the visual representations of an unlabeled dataset. Finally, you will explore methods for visualizing the features of a pretrained model on ImageNet, and also this model to implement Style Transfer.

The goals of this assignment are as follows:

- Understand and implement Transformer networks. Combine them with CNN networks for image captioning.
- Understand how to leverage self-supervised learning techniques to help with image classification tasks.
- Understand and implement techniques for image style transfer.

Q1: Image Captioning with Transformers (25 points)

The notebook `Transformer_Captioning.ipynb` will walk you through the implementation of a Transformer model and apply it to image captioning on COCO.

Q2: Self-Supervised Learning for Image Classification (15 points)

In the notebook `Self_Supervised_Learning.ipynb`, you will learn how to leverage self-supervised pretraining to obtain better performance on image classification tasks.

Q3: Style Transfer (10 points)

In the Jupyter notebook `StyleTransfer.ipynb` you will learn how to create images with the content of one image but the style of another.

Submitting your work:

Important. Please make sure that the submitted notebooks have been run and the cell outputs are visible.

Once you have completed all notebooks and filled out the necessary code, you need to follow the below instructions to submit your work:

To make sure everything is working properly, **remember to do a clean run ("Kernel -> Restart & Run All") after you finish work for each notebook** and submit the final version with all the outputs.

1. Generate a zip file of your code (`.py` and `.ipynb`) called `<UmassID>.zip` (For email address `arnaik@umass.edu` - zip file name is `arnaik.zip`). Please ensure you do not include the dataset folder and the `pretrained_model` folder in the zip.

If you run code on your local machine on Linux or macOS, you can run the provided `collectSubmission.sh` script from `assignment3/` to produce a file `<UmassID>.zip`.

2. Convert all notebooks (`.ipynb` files) into a single PDF file.

3. Please submit <UmassID>.zip and the pdf to Gradescope.